TEXORA · LECTURE SHEET

Measures of Dispersion

How far values scatter around the average — range, quartile deviation, mean deviation, variance and standard deviation, the coefficient of variation, and what the normal curve tells us about spread.

STAT-2109 · PROBABILITY AND STATISTICS · TOPIC 4 · LEVEL 2 TERM 1

1. Why averages are not enough

Two classes can share the same mean of 33 marks, one with every student between 28 and 38, the other split between failures near 10 and stars near 65. The averages agree; the classes are entirely different. A measure of **dispersion** (or variation) quantifies that scatter. **Absolute measures** express spread in the original units — marks, taka, centimetres. **Relative measures** divide an absolute measure by a suitable average, giving a pure number (often a percentage) that allows comparison between datasets in different units or of very different sizes. A good measure of dispersion, like a good average, should be rigidly defined, based on all observations, simple, algebraically workable and stable.

2. Range

The **range** is the simplest measure: the difference between the largest and smallest values.

$$\text{Range} = \text{Maximum} - \text{Minimum}$$

For the shared grouped dataset the classes span from the lowest boundary 0 to the highest boundary 70, so the range on class limits is $70 - 0 =$ **70 marks**. Its relative form, the coefficient of range, is (Maximum − Minimum)/(Maximum + Minimum). The range is quick and useful for a first look — and in quality control, where the spread of a small sample must be judged on the spot — but it rests on only two values, is violently affected by a single outlier, and says nothing about how the rest of the data are arranged between the extremes.

3. Quartile deviation

Quartiles (Topic 3) split the data into four equal parts. The **quartile deviation** (semi-interquartile range) is half the distance between the third and first quartiles:

$$\text{QD} = (Q_3 - Q_1)/2, \quad \text{Coefficient of QD} = (Q_3 - Q_1)/(Q_3 + Q_1)$$

Grouped quartiles use the same interpolation as the median: Q_1 in the class where cf reaches N/4, Q_3 where cf reaches 3N/4.

WORKED EXAMPLE

Shared dataset, N = 45. **Q_1:** N/4 = 11.25 falls in class 20–30 (cf 8 before, f = 10): $Q_1 = 20 + (11.25 - 8)/10 \times 10 =$ **23.25**. **Q_3:** 3N/4 = 33.75 falls in class 40–50 (cf 32 before, f = 8): $Q_3 = 40 + (33.75 - 32)/8 \times 10 =$ **42.19**. Therefore

$$\text{QD} = (42.19 - 23.25)/2 = 9.47 \text{ marks}; \quad \text{Coefficient of QD} = 18.94/65.44 = 0.289$$

The middle half of the class lies between 23.25 and 42.19 marks. Because QD uses only the middle 50%, it ignores the extremes completely — robust, but wasteful of information.

4. Mean deviation

The **mean deviation** (mean absolute deviation) averages the absolute distances of the values from a central point, usually the mean (or the median, which makes it smallest):

$$\text{MD (about mean)} = \Sigma f|x - \bar{x}| / N$$

WORKED EXAMPLE

Shared dataset about $\bar{x} = 33.0$. The $|x - 33|$ values for midpoints 5, 15, 25, 35, 45, 55, 65 are 28, 18, 8, 2, 12, 22, 32, and the $f|x - \bar{x}|$ terms are 56, 108, 80, 28, 96, 88, 32, summing to $\Sigma f|x - \bar{x}| = 488$. Hence

$$\text{MD} = 488 / 45 = 10.84 \text{ marks}$$

On average, a student's mark lies about 10.84 marks away from the class mean. The MD is intuitive and uses every observation, but absolute values resist further algebra, which is why the next measure displaced it.

5. Variance and standard deviation

Squaring the deviations instead of taking absolute values removes the sign problem and gives the most important measure in all of statistics. The **variance** is the mean of the squared deviations; the **standard deviation** (SD, written σ for a population) is its square root, restoring the original units:

$$\text{Variance} = \Sigma fx^2/N - \bar{x}^2, \quad \text{SD} = \sqrt{(\text{Variance})}$$

WORKED EXAMPLE

Shared dataset: the fx^2 column is 50, 1350, 6250, 17150, 16200, 12100, 4225, giving $\Sigma fx^2 = 57325$, and $\bar{x} = 33.0$. Then

$$\text{Variance} = 57325/45 - 33^2 = 1273.89 - 1089 = 184.89; \quad \text{SD} = \sqrt{184.89} = 13.60 \text{ marks}$$

The typical distance of a mark from the mean, in the root-mean-square sense, is 13.60 marks.

Properties of the standard deviation. (i) It is always non-negative, and zero only when all values are equal. (ii) Adding a constant to every value leaves the SD unchanged; multiplying every value by a constant k multiplies the SD by $|k|$ (and the variance by k^2). (iii) Variances of independent groups can be combined from their sizes, means and variances — a property no other dispersion measure offers so cleanly. (iv) For roughly bell-shaped data, most observations lie within two SDs of the mean (Section 7). These properties, plus its central role in probability, make the SD the standard measure of spread in research and industry.

6. Coefficient of variation: comparing consistency

The **coefficient of variation** (CV) expresses the SD as a percentage of the mean — the standard relative measure of dispersion:

$$\text{CV} = (\text{SD} / \text{Mean}) \times 100\%$$

For the shared dataset: CV = 13.60/33.0 × 100 = **41.2%**. The smaller the CV, the more consistent (less variable) the data.

WORKED EXAMPLE

Factory A: mean daily wage 500 taka, SD 40 → CV = 40/500 × 100 = **8%**. Factory B: mean 600 taka, SD 60 → CV = 60/600 × 100 = **10%**. Although B has the higher average wage, its wages are *less* consistent: the SD is larger both absolutely (60 > 40) and relative to its mean (10% > 8%). Factory A is the more consistent payer. Comparing SDs alone across different means — or different units — is a classic error; the CV exists to prevent it.

7. Spread under the normal curve

For data following the **normal distribution** (Topic on probability distributions), the SD acquires a precise geometric meaning. Measured in SD units from the mean μ, fixed proportions of the area — that is, of the observations — always fall within each band: about 68.27% within $\mu \pm 1\sigma$, 95.45% within $\mu \pm 2\sigma$ and 99.73% within $\mu \pm 3\sigma$. This is the basis of quality-control limits in industry and of the "six sigma" idea: a process whose specification limits lie six SDs from its mean produces almost no defectives.

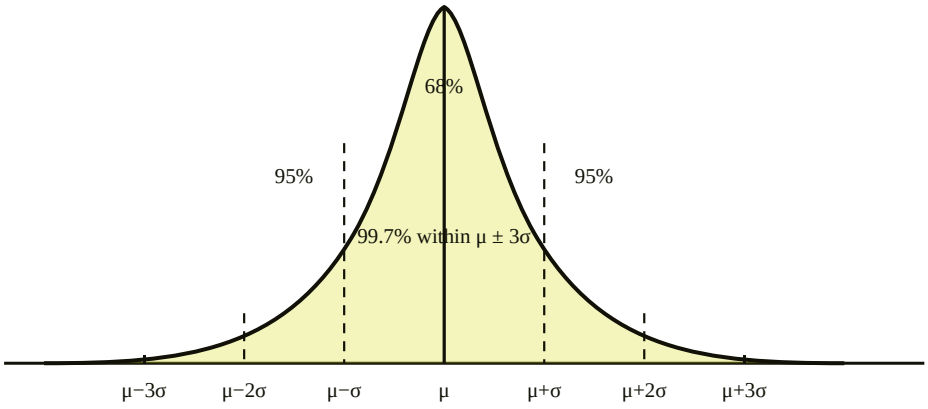


Figure 1 — The normal curve: about 68% of values lie within μ ± 1σ, 95% within μ ± 2σ and 99.7% within μ ± 3σ

8. The measures compared

MEASURE	BASED ON	STRENGTH	WEAKNESS
Range	Two extreme values only	Instant to compute; handy in quality control	Ruined by one outlier; ignores the interior
Quartile deviation	Middle 50% (Q_1, Q_3)	Unaffected by extremes; works with open-ended classes	Ignores half the data; weak algebraically
Mean deviation	All values, absolute deviations	Intuitive; uses every observation	Absolute values block further algebra
Standard deviation	All values, squared deviations	Best algebraic properties; basis of inference and control limits	Needs more computation; sensitive to extremes
Coefficient of variation	SD relative to the mean	Compares consistency across units and sizes	Meaningless if the mean is near zero

For the shared dataset the story is coherent: mean 33.0 marks, SD 13.60, MD 10.84, QD 9.47, range 70. Each measure in turn uses more of the data and rewards the effort with a fuller picture — and the CV of 41.2% records that this class's marks are, relative to their average, quite widely spread.

Glossary

TERM	MEANING

Dispersion	The scatter or spread of values around their average.
Absolute measure	A measure of spread expressed in the original units of the data.
Relative measure	A unit-free ratio of an absolute measure to an average, used for comparison.
Range	Largest value minus smallest value.
Interquartile range	$Q_3 - Q_1$: the spread of the middle half of the data.
Quartile deviation	Half the interquartile range, $(Q_3 - Q_1)/2$.
Mean deviation	The mean of the absolute deviations from the mean or median.
Variance	The mean of the squared deviations from the mean.
Standard deviation	The square root of the variance; the standard measure of spread.
Coefficient of variation	SD divided by the mean, times 100 — relative consistency in percent.

Key points & summary

- An average without a spread is half a description; dispersion completes it.
- Range of the shared data = 70 marks (class span 0 to 70) — quick, but based on two values only.
- $Q_1 = 23.25$ and $Q_3 = 42.19$ give QD = 9.47 and coefficient of QD = 0.289.
- Mean deviation about the mean = 488/45 = 10.84 marks.
- Variance = $\Sigma fx^2/N - \bar{x}^2 = 184.89$, so SD = 13.60 marks and CV = 41.2%.
- Compare consistency with the CV, never with raw SDs across different means: Factory A (CV 8%) beats Factory B (CV 10%).
- Under the normal curve: 68% within $\mu \pm 1\sigma$, 95% within $\mu \pm 2\sigma$, 99.7% within $\mu \pm 3\sigma$.

Practice

Answer in your own words; brief model answers follow.

1. Every value in a dataset is increased by 5, and then every value is doubled. State what happens to the SD at each step, and why.
2. Two sections have the same mean mark of 33. Section P has SD 6 and Section Q has SD 14. Which section is more consistent, and what extra fact does the SD give that the mean hides?
3. Yarn counts from two mills: Mill X, mean 40, SD 4; Mill Y, mean 25, SD 3. Which mill's yarn is more uniform? Show the calculation.

4. For a normal distribution with mean 100 and SD 15, between what two values do about 95% of the observations lie?
5. Explain why the quartile deviation is preferred to the range when a dataset contains one extreme outlier, and what information the QD throws away in exchange.
6. A grouped dataset has $\Sigma fx^2/N = 2000$ and mean 40. Find the variance, the SD and the CV.

Answer key

1. Adding 5 changes nothing: deviations from the mean are unchanged, so the SD is unchanged. Doubling every value doubles every deviation, so the SD doubles (and the variance quadruples).
2. Section P (SD 6) is far more consistent. The SD reveals the scatter: P's students are tightly clustered around 33, while Q's are widely spread, so identical means conceal very different classes.
3. CV of X = 4/40 × 100 = 10%; CV of Y = 3/25 × 100 = 12%. Mill X is more uniform, despite its larger SD, because relative to its mean its variation is smaller.
4. About 95% lie within $\mu \pm 2\sigma = 100 \pm 30$, that is, between 70 and 130.
5. The range is fixed entirely by the two extreme values, so one outlier inflates it without limit. The QD uses only Q_1 and Q_3, so extremes cannot affect it. In exchange, the QD ignores the top and bottom quarters of the data altogether — half the observations play no part.
6. Variance = $2000 - 40^2 = 2000 - 1600 = 400$. SD = $\sqrt{400} = 20$. CV = $20/40 \times 100 = 50\%$.

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